

🔬 Advanced Prompting Techniques

Day 3 — Notebook 1 of 4 | Estimated Time: 30 minutes

What You'll Learn

- Self-consistency: sample multiple reasoning paths and take the majority answer
- Tree-of-thought: explore multiple solution branches
- Iterative refinement: improve outputs through multi-step prompting

Setup

```
%pip install -U -q "google-genai>=1.0.0"
```

```
!pip install google-auth==2.47.0
```

```
Requirement already satisfied: google-auth==2.47.0 in /usr/local/lib/python3.12/dist-packages (2.47.0)  
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth==2.47.0) (0.4.0)  
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth==2.47.0) (4.9.1)  
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->google-auth==2.47.0) (0.6.1)
```

```
import sys, os  
import time  
  
import json  
import re  
from google import genai  
from google.genai import types  
from IPython.display import Markdown  
  
try:  
    from google.colab import auth  
    auth.authenticate_user()  
    print("✅ Authenticated via Colab")  
except ImportError:  
    print("✅ Using local credentials (gcloud auth)")  
  
client = genai.Client(  
    vertexai=True,  
    project="chanakya-476010",  
    location="us-central1",  
)  
MODEL_ID = "gemini-2.5-flash"  
print("✅ Ready!")
```

```
✅ Authenticated via Colab  
✅ Ready!
```

1. Self-Consistency

Idea: Ask the model the same reasoning question multiple times with higher temperature, then take the **majority answer**. This reduces errors on complex reasoning tasks.

```
Run 1: Think step by step → Answer: 42
Run 2: Think step by step → Answer: 42
Run 3: Think step by step → Answer: 38
Run 4: Think step by step → Answer: 42
Run 5: Think step by step → Answer: 42

Majority answer: 42 ✓ (4 out of 5)
```

```
from collections import Counter
import re

problem = """
A farmer has chickens and cows. He counts 30 heads and 80 legs total.
How many chickens does he have?

Think step by step. End your response with: ANSWER: [number]
"""

# Sample 5 reasoning paths
answers = []
for i in range(5):
    response = client.models.generate_content(
        model=MODEL_ID,
        contents=problem,
        config=types.GenerateContentConfig(temperature=0.7),
    )
    text = response.text
    # Extract the final answer
    match = re.search(r'ANSWER:\s*(\d+)', text)
    if match:
        answer = int(match.group(1))
        answers.append(answer)
        print(f"Run {i+1}: Answer = {answer}")
    else:
        print(f"Run {i+1}: Could not extract answer")

# Majority vote
if answers:
    counter = Counter(answers)
    majority = counter.most_common(1)[0]
    print(f"\n🏆 Self-consistent answer: {majority[0]} (appeared {majority[1]}/{len(answers)} times)")
```

```
Run 1: Answer = 20
Run 2: Answer = 20
Run 3: Answer = 20
Run 4: Answer = 20
Run 5: Answer = 20
```

```
🏆 Self-consistent answer: 20 (appeared 5/5 times)
```

2. Tree-of-Thought (ToT)

Idea: Instead of one linear chain of thought, explore **multiple solution branches**, evaluate each, and pick the best one.

```
prompt = """
Problem: A company needs to reduce costs by 20% without laying off employees.

Think about this using 3 different expert perspectives:

EXPERT 1 (Operations): Propose a solution focused on operational efficiency.
- Describe the approach
- List pros and cons
- Estimate savings
```

```
EXPERT 2 (Technology): Propose a solution focused on technology/automation.
- Describe the approach
- List pros and cons
- Estimate savings

EXPERT 3 (Finance): Propose a solution focused on financial restructuring.
- Describe the approach
- List pros and cons
- Estimate savings

FINAL VERDICT: After considering all three perspectives, which approach (or
combination) would you recommend and why?
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
Markdown(response.text)
```


EXPERT 1 (Operations): Propose a solution focused on operational efficiency.

Approach: The Operations expert would advocate for a comprehensive Lean Six Sigma initiative combined with supply chain optimization. The core idea is to identify and eliminate waste, streamline processes, and improve resource utilization across all departments, thereby increasing output and reducing the cost per unit of output without reducing the number of units or employees.

- 1. Process Mapping & Waste Elimination (Lean Principles):** Conduct detailed process mapping for key functions (e.g., procurement, production, customer service, administration). Identify non-value-added steps, bottlenecks, excessive waiting times, unnecessary movement, overproduction, defects, and over-processing (the 7 wastes of Lean). Implement process improvements to simplify, standardize, and accelerate workflows.
- 2. Supply Chain & Vendor Optimization:** Review all supplier contracts. Leverage bulk purchasing, renegotiate terms, consolidate vendors, explore alternative lower-cost suppliers, and optimize inventory management to reduce carrying costs and obsolescence.
- 3. Energy & Resource Efficiency:** Conduct energy audits. Implement energy-saving measures (e.g., LED lighting, smart HVAC systems, efficient machinery upgrades). Reduce consumption of office supplies, water, and other utilities.
- 4. Workforce Productivity & Cross-Training:** Maximize the productivity of existing employees by eliminating inefficiencies in their tasks. Implement cross-training programs to create a more flexible workforce, reducing idle time and improving response to fluctuating demand. Redeploy employees from redundant or optimized processes to areas of growth or higher value.

Pros:

- **Sustainable Savings:** Leads to permanent, structural cost reductions.
- **Improved Quality & Efficiency:** Often results in better product/service quality and faster delivery times.
- **Employee Engagement:** Involves employees in problem-solving, fostering a culture of continuous improvement.
- **No Layoffs:** Focuses on making existing employees more productive and effective.
- **Competitive Advantage:** A more efficient operation can lead to increased market share.

Cons:

- **Time & Resource Intensive:** Requires significant upfront effort, training, and commitment from management and employees.
- **Resistance to Change:** Employees may resist new processes or fear their jobs are at risk, despite the "no layoff" policy.
- **Complexity:** Identifying and implementing improvements across complex operations can be challenging.
- **Initial Investment:** Some efficiency upgrades (e.g., new machinery, energy systems) may require capital expenditure.
- **Gradual Results:** Significant savings may accrue over time rather than immediately.

Estimate Savings:

- **Process Optimization:** 5-10% of operational budget (by reducing errors, rework, processing time).
- **Supply Chain/Vendor Optimization:** 3-7% of Cost of Goods Sold (COGS) or relevant procurement spend.
- **Energy & Resource Efficiency:** 1-3% of total operating expenses (or 10-20% of utility bills).
- **Overall:** With a dedicated and aggressive program, operational efficiency alone could potentially achieve **8-15%** cost reduction, especially in companies with significant existing inefficiencies. Reaching 20% solely from this avenue would require a highly inefficient starting point and a very robust implementation.

EXPERT 2 (Technology): Propose a solution focused on technology/automation.

Approach: The Technology expert would propose leveraging automation, cloud solutions, and data analytics to reduce manual effort, optimize resource allocation, and enhance decision-making, thereby cutting costs without impacting employee numbers, but rather empowering them.

- 1. Robotic Process Automation (RPA):** Identify repetitive, rule-based tasks across finance, HR, customer service, and IT (e.g., data entry, invoice processing, report generation, system reconciliation). Implement RPA bots to perform these tasks, freeing up employees for higher-value, more strategic work.
- 2. Cloud Migration & SaaS Optimization:** Migrate on-premise IT infrastructure to cloud-based solutions (IaaS, PaaS) to reduce hardware, maintenance, and energy costs. Optimize Software-as-a-Service (SaaS) subscriptions by eliminating unused licenses, consolidating redundant tools, and negotiating better enterprise agreements.
- 3. AI & Machine Learning for Predictive Analytics:** Implement AI/ML tools for predictive inventory management (reducing carrying costs and waste), demand forecasting, preventative maintenance (reducing downtime and repair costs), and optimizing marketing spend.
- 4. Digital Collaboration & Remote Work Tools:** Enhance remote work capabilities with robust digital collaboration platforms. This can reduce office space requirements (e.g., hot-desking, reduced footprint), travel expenses, and associated utility costs.
- 5. Automated Customer Service & Support:** Deploy chatbots or AI-driven virtual assistants for routine customer inquiries, reducing the load on human customer service agents and allowing them to focus on complex issues.

Pros:

- **Significant Productivity Gains:** Automates tasks, leading to faster processing and higher output per employee.
- **Improved Accuracy & Reduced Errors:** Technology is less prone to human error, reducing rework and associated costs.
- **Scalability:** Technology solutions can often scale up or down more cost-effectively than human resources.
- **Data-Driven Insights:** Provides valuable data for further cost optimization and strategic decision-making.
- **Employee Upskilling:** Frees employees from mundane tasks, allowing them to be redeployed and upskilled into more strategic roles.

Cons:

- **High Upfront Investment:** Implementing new technology can require substantial capital outlay for software, hardware, and integration.
- **Implementation Complexity:** Integrating new systems with legacy infrastructure can be challenging and time-consuming.
- **Skill Gap:** Requires internal IT expertise or external consultants, which adds to cost. Employees need retraining for new roles.
- **Change Management:** Resistance from employees who fear technology will make their jobs obsolete (even with "no layoff" policy).
- **Security Risks:** Increased reliance on technology necessitates robust cybersecurity measures.

Estimate Savings:

- **RPA:** 5-15% of total operational costs (by automating a significant portion of white-collar tasks, freeing up employee time).
- **Cloud/SaaS Optimization:** 3-7% of IT budget (by reducing infrastructure and licensing costs).
- **Reduced Travel/Office Space:** 2-5% of administrative/overhead costs.
- **Overall:** Technology can be a powerful driver of cost reduction. With a focused strategy, savings could range from **10-20%**, particularly in organizations with high manual processes and outdated IT infrastructure.

EXPERT 3 (Finance): Propose a solution focused on financial restructuring.

Idea: Generate an initial output, then refine it through follow-up prompts.

Approach: The Finance expert would focus on optimizing the company's financial structure, scrutinizing all expenditures, and leveraging capital more

```
# Step 1: Generate initial draft
draft_prompt = "Write a product description for noise-cancelling wireless headphones priced at $199."
draft = client.models.generate_content(model=MODEL_ID, contents=draft_prompt)
print("📝 DRAFT:")
print(draft.text)
print()
```

4. Expense Freezes & Policy Adjustments: Implement a strict freeze on non-essential spending:

- Travel Freeze:** For non-essential travel, no existing travel arrangements. No new travel until a later date.
- Discretionary Spending Cuts:** Reduce budgets for travel, entertainment, conferences, non-critical training, and external consultants.
- Benefit Review:** Carefully review and potentially adjust non-salary employee benefits (e.g., reducing company-paid phone plans, adjusting health plan options – this must be handled *extremely* sensitively to avoid morale issues).

Escape the **Subscription Rationalization:** Cancel unused software subscriptions, news services, and other recurring fees.

5. Working Capital Optimization: Improve cash conversion cycle by accelerating accounts receivable collection, optimizing accounts payable terms (e.g., extending payment terms without penalty), and ensuring efficient cash management.

Powered by advanced **Active Noise Cancellation (ANC)** technology, the AuraNoise intelligently blocks out distractions – from r

Pros:

- Dynamic Sound That Moves You.**
- Immediate Impact:** Many financial adjustments (e.g., renegotiated contracts, expense freezes) can yield rapid cost reductions.
- Clear Metrics:** Cost savings are often easily quantifiable and directly impact the bottom line.
- Direct Control:** Financial decisions are often directly controlled by executive leadership.
- No Layoffs:** Primarily focuses on external spending and financial instruments, not internal headcount.

Say goodbye to tangled wires. Seamless Bluetooth 5.2 connectivity ensures a stable, high-fidelity connection to your devices

Cons:

- Morale Risk:** Cuts to employee benefits, travel, or discretionary spending can negatively impact morale and perceived value.
- Vendor Relationships:** Aggressive renegotiation can strain relationships with long-term suppliers.
- Short-Sightedness:** Can lead to cuts that hinder long-term growth or innovation if not carefully balanced.
- Limited Long-Term Strategic Value:** Doesn't fundamentally change how the company operates. Savings may plateau or not be sustainable without other initiatives.
- Extended Battery Life:** Enjoy up to 30 hours of playtime with ANC on (and even more without!), keeping your music going without other initiatives.
- Market Dependency:** Debt refinancing depends on prevailing interest rates.
- Intuitive On-Ear Controls:** Easily manage your music, calls, and voice assistant with a touch.

Estimate Savings:

- Vendor Renegotiation:** 5-10% off relevant charges spent you hours of playback when you're in a hurry.
- Debt Refinancing:** 1-3% reduction in interest expenses (if conditions are favorable).

You Expense Freeze/Dut: 3-6% of total operating expenses (by eliminating discretionary spending).

Whether you're commuting, working in the office, or enjoying your music, AuraNoise Headphones transform your listening

- Overall:** Financial restructuring can deliver substantial and relatively quick savings, potentially in the range of **7-18%**, depending on the initial spending habits and financial structure of the company.

Don't delay! Upgrade your audio experience with AuraNoise Wireless Noise-Cancelling Headphones today!*

FINAL VERDICT

```
# Step 2: Critique the draft
critique_prompt = f""""Review this product description and list 3 specific improvements:

<draft>
{draft.text}
</draft>

For each improvement, explain WHY it would make the description more effective."""""

critique = client.models.generate_content(model=MODEL_ID, contents=critique_prompt)
print("🔍 CRITIQUE:")
print(critique.text)
print()
```

streamline processes, and empower employees to be part of the solution. This will yield 8-10% over the medium term.

CRITIQUE:

Here are 3 specific improvements to your product description:

- 1. Targeted Technology & Automation:** Simultaneously, identify and implement key automation solutions (RPA) for high-volume, low-value tasks. Strategically migrate to cloud services where immediate ROI is clear. This can deliver an additional 5-10% in productivity gains and cost reduction.
- 2. Employee Redeployment & Upskilling (Crucial for "No Layoffs"):** The core of making this strategy work with no layoffs is a robust employee redeployment and upskilling program. As operational inefficiencies are removed and tasks are automated:
 - 1. Quantify "Rapid Changing" for greater impact.**
 - Employees freed from routine tasks must be trained and moved into higher-value activities: strategic projects, innovation, data analysis, customer experience enhancement, or supporting areas identified by operational reviews that need more hands-on attention.
 - 2. Current Text:** "Rapid Changing: A quick 10-minute change gives you hours of playback when you're in a hurry."
 - 3. Proposed change:** "A quick 10-minute change gives you hours of playback when you're in a hurry."
- 3. Justification for this Combination:**
 - Why it's effective:** The word "hours" is vague and can be interpreted broadly. By providing a concrete number (e.g., "3 h
 - Achieves the 20% Target:** By combining immediate financial cuts (e.g., 5-10%), with medium-to-long term operational efficiency (e.g., 8-15%) and targeted technological automation (e.g., 5-10%), the cumulative effect can comfortably exceed the 20% target.
 - Respects "No Layoffs" Constraint:** The emphasis is on increasing productivity per employee, redeploying talent, and investing in their future capabilities, rather than reducing headcount.

2. Enhance the description with a more vivid scenario and technical detail.

- Sustainability:** Operational and technological improvements create lasting reductions in operating costs and build a more efficient, agile company for the future, unlike purely financial cuts that may be short-lived.
- Resilience & Growth:** A more efficient, technologically advanced and financially disciplined organization is better positioned to weather future challenges and capitalize on new opportunities, fostering long-term growth.
- Mitigates Risks:** The phased approach allows for initial quick wins to build confidence, while the integrated nature ensures that financial cuts don't compromise the strategic investments needed for operational and technological transformation.

This integrated strategy demands strong leadership, but also communicates a clear commitment to the company's financial health and long-term strategic c and redeployment. It transforms the cost reduction challenge into an opportunity for strategic transformation and competitive advantage.

3. Add a section or bullet point about "Premium Build Quality" or "Durability."

```
* Current Text: The description focuses on sound, comfort (memory foam, lightweight), and features, but doesn't explicitly
* Proposed Change (Add as a new bullet point in "Key Features You'll Love"):
*   "Premium & Durable Build:" Constructed with high-grade, robust materials and a sleek, modern finish, these headph
* Why it's effective: For headphones priced at $199 and aimed at "discerning listeners," durability and material quality a
```

```
# Step 3: Apply improvements
refine_prompt = f"""Rewrite this product description by applying these improvements:
```

```
<original>
{draft.text}
</original>
```

```
<improvements>
{critique.text}
</improvements>
```

```
Write the improved version:"""
```

```
refined = client.models.generate_content(model=MODEL_ID, contents=refine_prompt)
print(" 🌟 REFINED:")
Markdown(refined.text)
```

🌟 REFINED:

Immerse Yourself in Pure Sound: Introducing the [Brand Name] AuraNoise Headphones

Price: \$199.00

Escape the chaos and dive into a world of pristine audio with the **[Brand Name] AuraNoise Wireless Noise-Cancelling Headphones**. Engineered for discerning listeners who demand both exceptional sound and unparalleled tranquility, the AuraNoise is your personal sanctuary in a noisy world.

Silence the World, Elevate Your Audio. Powered by advanced **Active Noise Cancellation (ANC)** technology, the AuraNoise intelligently blocks out distractions – from roaring airplane engines to bustling city streets. Create your own private acoustic space, allowing you to focus on your music, podcasts, or work without interruption.

Dynamic Sound That Moves You. Experience your favorite tracks like never before. Equipped with precision-engineered 40mm drivers, the AuraNoise delivers **rich, dynamic audio** with deep, resonant bass, crystal-clear mids, and crisp, articulate highs. Every note, every beat, every word comes through with stunning clarity and immersive depth.

Untethered Freedom, All-Day Comfort. Say goodbye to tangled wires. **Seamless Bluetooth 5.2 connectivity** ensures a stable, high-fidelity connection to your devices up to 33 feet away. Designed for marathon listening sessions, the AuraNoise features **plush, memory foam earcups** and an adjustable, lightweight headband that conforms gently to your head, providing cloud-like comfort for hours on end.

Key Features You'll Love:

- **Superior Active Noise Cancellation (ANC):** Block out distractions and find your focus.
- **Immersive Audio:** 40mm drivers deliver rich bass, clear mids, and detailed highs.
- **Premium & Durable Build:** Constructed with **high-grade, robust materials** and a sleek, modern finish, these headphones are built to withstand daily use while maintaining their aesthetic appeal and comfortable fit for years to come.
- **Extended Battery Life:** Enjoy **up to 30 hours of playtime** with ANC on (and even more without!), keeping your music going on the longest journeys.
- **Crystal-Clear Calls:** Featuring an advanced integrated microphone with **Qualcomm cVc™ 8.0 Noise Reduction**, your voice is precisely isolated from surrounding chatter, ensuring crystal-clear conversations even on a bustling city street or during a busy commute. Make important calls with confidence, anywhere.
- **Intuitive On-Ear Controls:** Easily manage your music, calls, and voice assistant with a touch.
- **Ergonomic & Portable Design:** Foldable earcups and an included travel case make them the perfect companion for your commute or travels.
- **Rapid Charging:** A quick 10-minute charge delivers **3 hours of playback**, perfect for when you're short on time.

Your Soundtrack, Uninterrupted. Whether you're commuting, working, studying, or simply unwinding, the [Brand Name] AuraNoise Headphones transform your listening experience. At **\$199.00**, they offer premium features and performance that elevate your everyday.

Don't just listen; immerse. Order your AuraNoise Wireless Noise-Cancelling Headphones today!

📌 Exercise 1: Self-Consistency on a Logic Puzzle

Use self-consistency (5 samples) to solve this logic puzzle:

```
# Exercise 1: Self-consistency
# TODO: Run this 5 times and take the majority answer

puzzle = """
In a race, you overtake the person in second place.
What place are you in now?

Think step by step. End with ANSWER: [your answer]
"""

# TODO: Implement self-consistency sampling
# Hint: Use the pattern from Section 1 above
```